Chapter 13A
Integrated Systems, Diagnostic Skills, and Final Internal Medicine Review

Medvale Internal Medicine - capstone chapter for cross-system reasoning, bedside interpretation, statistics, diagnostic algorithms, and final synthesis.

Purpose of this final chapter
Organ-system chapters teach diseases; practice often begins with unstable vital signs, a chief complaint, an abnormal lab, or an ambiguous outpatient presentation. This chapter converts prior content into a cross-system diagnostic map. It is intentionally paragraph-heavy and algorithm-rich so that each symptom, sign, test, and disease relationship can be extracted into a future graph-RAG structure.

1 Orientation: how to think across systems

A final internal medicine review should not merely list diseases again. The core skill is translation: converting a patient presentation into a problem representation, matching that representation to illness scripts, choosing tests that move probability, and acting before certainty when the cost of delay is high. The most dangerous errors occur when the clinician treats an organ system as isolated. Dyspnea may be pulmonary edema, pulmonary embolism, pneumonia, anemia, acidosis, neuromuscular weakness, anxiety, or a medication effect. Altered mental status may be sepsis, hypoglycemia, hypercapnia, stroke, seizure, hepatic encephalopathy, uremia, intoxication, withdrawal, or delirium from polypharmacy. The final review therefore emphasizes syndromes and thresholds rather than isolated facts.

For board-style reasoning and clinical reasoning alike, every presentation can be processed through four questions. First, is the patient unstable or at risk of immediate harm? Second, what syndrome best summarizes the presentation? Third, what few diagnoses explain the most dangerous and most likely possibilities? Fourth, which test or treatment must occur now because delay changes outcome? The answer may be a diagnostic test, but it may also be empiric antibiotics, anticoagulation, dextrose, epinephrine, oxygen, fluids, vasopressors, or urgent consultation.

Clinical layer	Key question	Examples	Common trap
Stability	Is there airway, breathing, circulation, neurologic, or perfusion failure?	Hypotension, severe hypoxemia, obtundation, active GI bleed, STEMI, stroke window, septic shock.	Continuing outpatient-style differential diagnosis while the patient needs resuscitation.
Syndrome	What pattern summarizes the case?	Pleuritic chest pain plus hypoxemia; fever plus murmur plus emboli; AKI plus hematuria plus casts; rash plus mucosal lesions.	Anchoring on the first abnormal lab without building the syndrome.
Probability	Which diseases are likely and which are dangerous?	Migraine likely but SAH dangerous; viral URI likely but epiglottitis dangerous; GERD likely but ACS dangerous.	Testing only for common diagnoses or only for rare diagnoses.
Action threshold	What must be treated before proof?	Epinephrine for anaphylaxis; antibiotics for meningitis/sepsis; steroids for GCA vision threat; aspirin/reperfusion for ACS.	Waiting for confirmatory tests when disease has high time-dependent morbidity.
Disposition	Can this be safely managed outpatient?	Stable cellulitis vs necrotizing infection; low-risk syncope vs arrhythmia; uncomplicated UTI vs pyelonephritis/sepsis.	Assuming normal current vitals equal low future risk.

2 Diagnostic reasoning framework

2.1 Problem representation

A problem representation is a concise one-sentence abstraction that includes age, relevant risk factors, tempo, key symptoms, objective findings, and localization. It should not simply repeat the chart. The purpose is to make the differential easier to search mentally. For example, “62-year-old smoker with abrupt pleuritic chest pain, tachycardia, hypoxemia, and unilateral leg swelling” is more useful than “chest pain and shortness of breath.” The first sentence points toward pulmonary embolism while preserving alternative diagnoses such as pneumothorax, pneumonia, ACS, and aortic catastrophe.

Good problem representations use semantic qualifiers: acute vs chronic, painful vs painless, inflammatory vs noninflammatory, focal vs diffuse, compensated vs decompensated, volume-depleted vs overloaded, obstructive vs restrictive, hepatocellular vs cholestatic, nephritic vs nephrotic, central vs peripheral, upper vs lower motor neuron, and infectious vs autoimmune vs malignant. These qualifiers become graph edges linking presentations to disease families.

Raw finding	Semantic qualifier	Differential consequence
Fever for 2 days with cough	Acute infectious respiratory syndrome	Viral URI, influenza/COVID, pneumonia, bronchitis, asthma/COPD exacerbation with trigger.
Hematuria plus RBC casts plus hypertension	Nephritic renal syndrome	Glomerulonephritis, vasculitis, lupus nephritis, postinfectious GN, IgA nephropathy.
Painless jaundice plus weight loss	Cholestatic/obstructive malignant pattern	Pancreatic cancer, cholangiocarcinoma, ampullary tumor, less often gallstone obstruction.
Symmetric MCP/PIP swelling with morning stiffness	Inflammatory small-joint polyarthritis	RA, SLE, viral arthritis, psoriatic arthritis, mixed connective tissue disease.
Vertigo for seconds triggered by head motion	Brief triggered peripheral vestibular syndrome	BPPV more likely than vestibular neuritis, stroke, migraine.
Acute dyspnea with wheezing and hypercapnia	Obstructive ventilatory failure	Asthma/COPD exacerbation, anaphylaxis, airway obstruction, medication effect.

2.2 Illness scripts and discriminating features

An illness script is a compact disease model: predisposing conditions, pathophysiology, time course, key symptoms, signs, labs, complications, and treatment. In final review, the most useful part of an illness script is the discriminating feature - the finding that separates two similar diagnoses. Pleuritic chest pain occurs in PE, pneumonia, pneumothorax, pericarditis, and rib disease; unilateral leg swelling, tachycardia, and unexplained hypoxemia shift the script toward PE. Fever and focal crackles shift toward pneumonia. Sudden unilateral absent breath sounds shift toward pneumothorax. Positional pain relieved by leaning forward shifts toward pericarditis.

Similar presentations	Discriminating features
ACS vs GERD vs pericarditis	ACS: exertional pressure, diaphoresis, ECG/troponin changes. GERD: burning/regurgitation, postprandial, response to acid suppression. Pericarditis: pleuritic positional pain, diffuse ST elevation/PR depression, friction rub.
Pneumonia vs COPD/asthma exacerbation	Pneumonia: fever, focal crackles, infiltrate. COPD/asthma: wheeze, prolonged expiration, hyperinflation, no focal infiltrate unless trigger infection.
Migraine vs subarachnoid hemorrhage	Migraine: recurrent stereotyped attacks, photophobia, nausea, gradual evolution. SAH: thunderclap onset, worst headache, meningismus, exertional onset, neurologic deficit.
Delirium vs dementia vs depression	Delirium: acute/fluctuating attention impairment. Dementia: chronic progressive cognitive decline. Depression: low mood/anhedonia with cognitive complaints; may have intact effort-dependent testing.
Iron deficiency anemia vs anemia of chronic inflammation	Iron deficiency: low ferritin, high TIBC, microcytosis, blood loss. Chronic inflammation: low serum iron with low/normal TIBC and normal/high ferritin.
Nephrotic vs nephritic syndrome	Nephrotic: heavy proteinuria, edema, hypoalbuminemia, hyperlipidemia. Nephritic: hematuria, RBC casts, hypertension, reduced GFR, variable proteinuria.

2.3 Bayesian reasoning and test interpretation

Tests should be ordered because the result will change probability enough to change action. Sensitivity and specificity describe the test in diseased and nondiseased populations; predictive values describe what a positive or negative result means in the patient in front of you and therefore depend heavily on pretest probability. A highly sensitive test is useful for ruling out disease when negative, but only when the pretest probability is not already very high. A highly specific test is useful for ruling in disease when positive, but false positives become common when testing is applied indiscriminately to low-risk patients.

A classic example is D-dimer. In a low-risk patient with possible PE, a negative high-sensitivity D-dimer can reduce post-test probability enough to avoid CT pulmonary angiography. In a high-risk patient with hypoxemia, tachycardia, and unilateral leg swelling, D-dimer is not the decisive test because the pretest probability is high; imaging and/or empiric anticoagulation may be needed depending on stability and bleeding risk. The same principle applies to troponin, ANA, Lyme serologies, urine cultures, thyroid testing, and tumor markers.

Concept	Formula/meaning	Clinical interpretation
Sensitivity	True positive / all with disease	High sensitivity helps rule out when negative if pretest probability is low or moderate.
Specificity	True negative / all without disease	High specificity helps rule in when positive, especially if the result matches the syndrome.
Positive predictive value	Probability of disease given positive test	Rises as disease prevalence/pretest probability rises. Low in indiscriminate screening.
Negative predictive value	Probability of no disease given negative test	Falls when pretest probability is high. A negative test may not be enough in a classic presentation.
Likelihood ratio positive	Sensitivity / (1 - specificity)	LR+ greater than 10 often produces a large probability increase; 2 to 5 is modest.
Likelihood ratio negative	(1 - sensitivity) / specificity	LR- less than 0.1 often produces a large probability decrease; 0.2 to 0.5 is modest.
Spectrum bias	Test performance changes with disease severity/population	A test validated in severe inpatient disease may perform differently in mild outpatient disease.

Testing rule for the final review
Before ordering a test, ask: What disease am I testing for? What is the pretest probability? What result would change management? What are the harms of a false positive, false negative, incidentaloma, contrast exposure, radiation, cost, or delay?

3 Cross-system chief complaint algorithms

3.1 Chest pain

Chest pain is first a stability problem, not a diagnostic puzzle. Immediate concerns are ACS, aortic dissection, pulmonary embolism, tension pneumothorax, esophageal rupture, pericardial tamponade, and severe pneumonia/sepsis. Character, location, radiation, exertional component, pleuritic quality, positional features, associated dyspnea/syncope/diaphoresis, neurologic deficits, pulse deficits, and risk factors shape the workup. ECG and troponin are central for possible ACS, but a normal initial ECG does not fully exclude unstable angina or early MI. Severe tearing pain to the back with pulse deficit or neurologic symptoms should trigger concern for dissection. Pleuritic pain with hypoxemia and tachycardia should trigger PE or pneumothorax evaluation.

Chest pain algorithm

1. Assess ABCs, vital signs, oxygenation, perfusion, mental status, and obtain immediate ECG if cardiac ischemia is possible.
2. If STEMI or high-risk ACS pattern is present, activate reperfusion pathway and give antiplatelet/anticoagulation therapy as appropriate unless contraindicated.
3. If widened mediastinum, pulse deficit, focal neurologic deficit, abrupt tearing pain, or severe hypertension suggests dissection, avoid reflex anticoagulation until dissection is evaluated.
4. If pleuritic pain, hypoxemia, tachycardia, hemoptysis, recent surgery/immobility, cancer, pregnancy/postpartum, or unilateral leg swelling suggests PE, use risk stratification to decide D-dimer vs imaging.
5. If unilateral absent breath sounds or sudden pleuritic pain suggests pneumothorax, obtain urgent imaging unless tension physiology requires immediate decompression.
6. If stable and dangerous causes are unlikely, consider GERD, esophageal spasm, costochondritis, anxiety/panic, zoster prodrome, pneumonia, or pericarditis.

Diagnosis	Clues	Initial tests/actions
ACS	Pressure, exertional, diaphoresis, radiation, risk factors, abnormal ECG/troponin.	ECG within minutes, serial troponins, aspirin, risk stratification, reperfusion/anticoagulation when indicated.
Aortic dissection	Abrupt severe tearing pain, pulse/BP differential, neurologic deficit, aortic regurgitation, connective tissue disease.	CTA chest/abdomen if stable; beta-blockade and BP control; surgical consult for type A.
Pulmonary embolism	Pleuritic pain, dyspnea, tachycardia, hypoxemia, leg swelling, provoking risk.	Risk score, D-dimer if low/intermediate, CTPA/VQ, anticoagulation if likely and safe.
Pericarditis	Sharp pleuritic positional pain, friction rub, diffuse ST elevation/PR depression.	ECG, troponin, echo if effusion/tamponade concern, NSAID/colchicine if uncomplicated.
Pneumothorax	Sudden pleuritic pain, unilateral decreased breath sounds, tall thin/COPD/trauma.	CXR/ultrasound; needle decompression if tension physiology.
Esophageal rupture	Severe pain after vomiting/instrumentation, subcutaneous emphysema, sepsis.	CT/esophagram, NPO, broad antibiotics, surgery.

3.2 Dyspnea

Dyspnea can arise from oxygenation failure, ventilation failure, increased work of breathing, low oxygen carrying capacity, metabolic acidosis, cardiac dysfunction, neuromuscular weakness, or anxiety. The first distinction is whether the patient is hypoxemic, hypercapnic, acidotic, obstructed, overloaded, infected, embolic, or weak. Pulse oximetry can be misleading in carbon monoxide poisoning, severe anemia, poor perfusion, or methemoglobinemia. ABG or VBG helps identify hypercapnia and acid-base stress. CXR, ECG, BNP, troponin, bedside ultrasound, and PE testing are chosen based on syndrome.

Acute dyspnea algorithm

1. Assess work of breathing, oxygen saturation, mental status, airway patency, and hemodynamics. Give oxygen or ventilatory support based on severity.
2. Look for immediate reversible killers: anaphylaxis, tension pneumothorax, airway obstruction, pulmonary edema, severe asthma/COPD, PE, pneumonia/sepsis, ACS, arrhythmia.
3. Use exam and bedside data to classify: wheezing/obstruction, crackles/edema, focal findings/infection, pleuritic/hypoxemic/PE, clear lungs/metabolic acidosis/anemia/neuromuscular.
4. Obtain targeted tests: CXR, ECG, VBG/ABG, CBC, BMP, troponin/BNP, viral testing, D-dimer/CTPA when appropriate.
5. Treat the syndrome while confirming: bronchodilators/steroids for obstructive flare, diuretics/nitrates for pulmonary edema, antibiotics for pneumonia, anticoagulation for PE if likely and safe.

Pattern	Likely mechanisms	Pearls
Hypoxemia with focal infiltrate	Pneumonia, aspiration, pulmonary hemorrhage, infarct.	Fever and sputum support infection; hemoptysis/anemia/renal disease suggests hemorrhage.
Hypoxemia with clear CXR	PE, early pneumonia, asthma/COPD, shunt, pulmonary hypertension.	Do not dismiss severe dyspnea because CXR is normal.
Wheezing	Asthma/COPD, anaphylaxis, CHF/cardiac asthma, foreign body.	Wheezing is airflow limitation, not automatically asthma.
Crackles plus edema/JVP	Heart failure, renal failure, volume overload.	Bedside ultrasound B-lines and IVC can support congestion.
Kussmaul respirations	Metabolic acidosis: DKA, uremia, lactic acidosis, toxins.	Oxygen saturation may be normal while the patient is critically ill.
Weak cough plus shallow breathing	Neuromuscular weakness, cervical cord disease, myasthenia, Guillain-Barre.	Monitor negative inspiratory force/vital capacity; intubate before crash.

3.3 Syncope

Syncope is transient loss of consciousness due to global cerebral hypoperfusion with rapid recovery. The main task is not to prove vasovagal syncope; it is to identify cardiac syncope, arrhythmia, structural heart disease, pulmonary embolism, hemorrhage, seizure mimics, and orthostatic hypotension. High-risk features include syncope during exertion, syncope while supine, palpitations before syncope, family history of sudden death, known structural heart disease, abnormal ECG, persistent hypotension, severe anemia/bleeding, chest pain, dyspnea, or neurologic deficit. Routine neuroimaging is low yield when the history is classic syncope without focal neurologic findings.

Type	Clinical clues	Evaluation focus
Vasovagal/reflex	Trigger, prodrome, nausea, warmth, diaphoresis, prolonged standing, rapid recovery.	History; avoid overtesting if no red flags.
Orthostatic	Occurs after standing, dehydration, bleeding, autonomic failure, medication effect.	Orthostatic vitals, volume status, medication review, CBC/BMP if indicated.
Cardiac arrhythmia	Sudden without prodrome, exertional/supine, palpitations, abnormal ECG.	ECG, telemetry/Holter/event monitor, echo if structural concern.
Structural cardiopulmonary	Exertional syncope, murmur, heart failure, PE symptoms.	Echo, PE evaluation, cardiology depending on findings.
Seizure mimic	Tongue biting, prolonged postictal state, witnessed tonic-clonic activity.	Glucose/electrolytes; EEG/neuro only when history supports seizure.

3.4 Altered mental status

Altered mental status is a syndrome of brain dysfunction caused by systemic, neurologic, toxic, metabolic, infectious, endocrine, respiratory, or medication-related disease. The first test is bedside glucose. The first actions are airway assessment, oxygenation/ventilation, circulation, temperature, medication/toxin review, and neurologic screen. Delirium is acute and fluctuating with impaired attention; dementia is chronic cognitive decline; psychiatric illness should be diagnosed only after physiologic causes are considered. A normal CT head does not exclude meningitis, encephalitis, nonconvulsive seizure, intoxication, hepatic encephalopathy, or metabolic disease.

AMS algorithm

1. Check airway, breathing, circulation, oxygen saturation, fingerstick glucose, temperature, and medication/toxin exposure immediately.
2. Treat reversible emergencies: dextrose for hypoglycemia; naloxone for opioid toxidrome with respiratory depression; thiamine before glucose if Wernicke risk and feasible; cooling/heating for temperature extremes.
3. Look for focal neurologic deficit, meningismus, seizure activity, trauma, papilledema, immunosuppression, anticoagulation, or severe hypertension.
4. Obtain targeted labs: CBC, CMP, calcium, sodium, renal/liver tests, ABG/VBG if hypercapnia/acidosis suspected, UA/cultures if infection likely, toxicology/levels when indicated.
5. Use CT/LP/EEG based on syndrome: CT for trauma/focal deficit/anticoagulation/papilledema; LP for meningitis/encephalitis after safety screen; EEG for nonconvulsive status or unexplained persistent AMS.

3.5 Fever, FUO, and sepsis

Fever is not synonymous with infection. The differential includes infection, malignancy, autoimmune/autoinflammatory disease, thromboembolism, drug fever, endocrine disease, transfusion reactions, and environmental heat illness. In unstable patients, treat sepsis early while obtaining cultures when this does not delay antibiotics. In stable patients, the pattern and duration matter. Fever of unknown origin requires repeated history and exam, medication review, travel/exposure assessment, HIV/TB risk assessment, inflammatory markers, cultures, age-appropriate malignancy evaluation, and targeted imaging rather than random testing.

Fever context	High-yield differential	Initial direction
Fever plus hypotension/lactate/organ dysfunction	Sepsis, hemorrhage with inflammatory response, adrenal crisis, anaphylaxis, pancreatitis.	Resuscitation, cultures if feasible, broad antibiotics if infection likely, source control.

Fever plus new murmur/emboli/IVDU/prosthetic valve	Endocarditis.	Multiple blood cultures, echocardiography, targeted antibiotics after cultures unless unstable.
Fever plus headache/neck stiffness/AMS	Meningitis, encephalitis, SAH mimic.	Empiric antibiotics/acyclovir as indicated; CT before LP if safety criteria.
Fever plus rash plus hypotension	Toxic shock, meningococcemia, RMSF, SJS/TEN with sepsis, DRESS.	Isolation/empiric therapy, skin/mucosal exam, medication history.
Prolonged fever plus weight loss/night sweats	TB, lymphoma, endocarditis, abscess, autoimmune disease.	Repeat exam, cultures, inflammatory markers, HIV/TB testing, imaging guided by clues.
Fever after new medication	Drug fever, DRESS, serotonin syndrome, NMS, malignant hyperthermia.	Medication timeline, eosinophils/LFTs, rigidity/clonus/autonomic signs.

3.6 Shock

Shock is inadequate tissue perfusion. Hypotension may be absent early, especially in younger patients or those with strong sympathetic tone. Classify shock physiologically: distributive, hypovolemic, cardiogenic, obstructive, or mixed. Skin findings, JVP, lung exam, pulse pressure, bedside ultrasound, lactate, urine output, ECG, CXR, and response to fluids narrow the diagnosis. Septic shock can coexist with cardiomyopathy, hemorrhage can coexist with anticoagulation, and pulmonary embolism can cause obstructive shock with hypoxemia and right ventricular strain.

Shock type	Mechanism	Clues	Initial management concept
Distributive	Low systemic vascular resistance.	Warm early sepsis, fever, wide pulse pressure; anaphylaxis with urticaria/wheeze; neurogenic with bradycardia.	Fluids, vasopressors, source-specific therapy: antibiotics/source control or epinephrine.
Hypovolemic	Low preload from blood/fluid loss.	Flat neck veins, dry mucosa, bleeding, diarrhea/vomiting, high BUN/Cr ratio.	Volume/blood resuscitation; stop losses; source control.
Cardiogenic	Pump failure.	JVP, crackles, cool extremities, S3, ischemic ECG, low EF.	Avoid indiscriminate fluids; inotropes/vasopressors, revascularization, diuretics/ventilation depending on congestion.
Obstructive	Blocked filling/outflow.	Tamponade JVP/muffled sounds; tension pneumothorax; massive PE with RV strain.	Procedure or targeted therapy: pericardiocentesis, decompression, thrombolysis/embolectomy when indicated.
Mixed	Multiple mechanisms.	Sepsis plus cardiomyopathy; trauma plus tamponade; pancreatitis plus third spacing.	Treat dominant physiology while reassessing frequently.

3.7 Other cross-system patterns

Unintentional weight loss is concerning when progressive, objectively documented, or associated with fever, drenching night sweats, lymphadenopathy, anemia, hypercalcemia, abnormal liver tests, GI bleeding, dysphagia, or age-appropriate cancer risk. Lymph nodes are interpreted by size, location, tenderness, consistency, fixation, duration, and systemic symptoms. Supraclavicular, hard/fixed, generalized, or persistent unexplained nodes warrant more aggressive evaluation. Edema is classified by hydrostatic pressure, oncotic pressure, permeability, lymphatic obstruction, and sodium retention; unilateral painful edema should trigger DVT/cellulitis/trauma considerations, while bilateral edema points toward heart, kidney, liver, venous, endocrine, medication, or nutritional causes.

Abdominal pain is localized by quadrant but prioritized by instability, peritonitis, vascular catastrophe, pregnancy risk, immunosuppression, age, and surgical history. Pain out of proportion with atrial fibrillation or lactic acidosis suggests mesenteric ischemia. GI bleeding is approached by hemodynamic status, upper vs lower source, anticoagulant use, liver disease, and need for transfusion/endoscopy. Abnormal labs require context: hyperkalemia is triaged by ECG and severity, hyponatremia by symptoms and acuity, hypercalcemia by symptoms/PTH/malignancy, and AKI by prerenal, intrinsic, postrenal, or mixed mechanisms. Rash with fever, hypotension, mucosal involvement, skin pain, necrosis, purpura, facial edema, eosinophilia, or organ injury is urgent.

Presentation	Do-not-miss diagnoses	Key clues
Weight loss plus night sweats	Lymphoma, TB, HIV, endocarditis, malignancy, autoimmune disease.	B symptoms, lymphadenopathy, exposures, anemia, hypercalcemia.
Bilateral edema	Heart failure, renal disease, cirrhosis, venous disease, medications, hypothyroidism.	JVP, urine protein, albumin, LFTs, creatinine, medication list.
Abdominal pain out of proportion	Mesenteric ischemia.	Atrial fibrillation, vascular disease, lactate may be late; CTA.
Upper GI bleeding	PUD, varices, gastritis/esophagitis, malignancy, Mallory-Weiss.	Hemodynamics first; IV PPI; octreotide/antibiotics if variceal risk.
AKI plus hematuria/casts	Glomerulonephritis, vasculitis, lupus nephritis, anti-GBM.	Urinalysis microscopy, protein quantification, complements/serologies, nephrology.
Rash plus mucosal lesions	SJS/TEN, erythema multiforme major, autoimmune blistering disease.	Stop culprit drug; admit/derm/ophtho if severe.

4 ECG interpretation

ECG interpretation should be systematic. Rate and rhythm come first, then axis, intervals, chamber enlargement, ischemia/infarction, conduction blocks, electrolyte/drug effects, and comparison with prior tracings. The goal is not to memorize every pattern in isolation but to connect ECG findings to clinical syndromes. Chest pain with ST elevation is different from diffuse ST elevation in pericarditis; syncope with bifascicular block is different from asymptomatic conduction delay; hyperkalemia with peaked T waves demands treatment even before the full cause is known.

Systematic ECG read

1. Confirm patient, calibration, and technical quality. Look for artifact, lead reversal, and baseline wander.
2. Rate: estimate ventricular rate using R-R interval or large boxes; identify bradycardia, normal rate, tachycardia.
3. Rhythm: decide if regular or irregular; look for P waves before every QRS; determine sinus rhythm, AF, flutter, SVT, VT, or paced rhythm.
4. Axis: use leads I and aVF for quick quadrant; left axis may suggest LAFB/LVH/inferior MI; right axis may suggest RV strain, LPFB, COPD, or lateral MI.
5. Intervals: PR, QRS, QTc. Prolonged PR suggests AV nodal delay; wide QRS suggests bundle branch block, ventricular rhythm, hyperkalemia, sodium-channel blockade, or preexcitation.
6. Ischemia/infarction: inspect ST elevation/depression, T-wave inversion, Q waves, reciprocal changes, and anatomic territories.
7. Look for special patterns: hyperkalemia, hypokalemia, hypocalcemia, digoxin effect/toxicity, pericarditis, PE/RV strain, Brugada pattern, WPW.

ECG pattern	Major associations	Clinical action idea
ST elevation in contiguous leads with reciprocal changes	STEMI/acute coronary occlusion.	Reperfusion pathway if clinical context fits.
Diffuse concave ST elevation plus PR depression	Acute pericarditis.	Assess for myocarditis/tamponade; NSAID/colchicine if uncomplicated.
ST depression V1-V3 with tall R waves	Posterior MI.	Posterior leads; manage as MI if confirmed/suspected.
Peaked T waves, PR prolongation, QRS widening	Hyperkalemia.	Calcium for ECG changes/severe K; shift and remove potassium.

U waves, ST depression, flattened T waves	Hypokalemia.	Replace K/Mg; evaluate diuretic/GI losses.
Irregularly irregular rhythm without P waves	Atrial fibrillation.	Rate/rhythm decision, anticoagulation risk assessment, trigger evaluation.
Wide-complex regular tachycardia	VT until proven otherwise.	If unstable cardiovert; avoid assuming SVT with aberrancy.
Delta wave plus short PR	WPW/preexcitation.	Avoid AV nodal blockers in preexcited AF.

5 Radiographic interpretation

A chest radiograph should be read in a fixed order so that subtle findings are not missed. First check film quality: patient identity, date, projection, rotation, inspiration, exposure, and portable status. Then inspect airway/mediastinum, bones/soft tissues, cardiac silhouette, diaphragms, pleura, lungs, devices/lines, and hidden areas behind the heart and below the diaphragms. Portable AP films exaggerate heart size and can obscure posterior effusions. A normal CXR does not exclude PE, early pneumonia, asthma/COPD exacerbation, ACS, or metabolic dyspnea.

CXR finding	Interpretation	Common clinical links
Focal lobar consolidation	Airspace filling.	Bacterial pneumonia, aspiration, pulmonary hemorrhage, infarct.
Diffuse bilateral interstitial/alveolar opacities	Edema, ARDS, viral/atypical infection, hemorrhage.	Use heart size, effusions, volume status, BNP, echo, hemoglobin/renal findings.
Hyperinflation/flattened diaphragms	Air trapping.	COPD/asthma; may coexist with pneumonia or CHF.
Pleural effusion	Fluid in pleural space.	CHF, malignancy, pneumonia/empyema, PE, cirrhosis, nephrotic syndrome.
Pneumothorax	Pleural line without lung markings beyond.	Primary spontaneous, COPD, trauma, procedures; tension if mediastinal shift/hypotension.
Free air under diaphragm	Pneumoperitoneum.	Perforated viscus until proven otherwise.

Abdominal imaging is chosen by the suspected process. RUQ ultrasound is first-line for biliary colic/cholecystitis. CT abdomen/pelvis is useful for appendicitis, diverticulitis, obstruction, perforation, abscess, pancreatitis complications, stones when noncontrast protocol is used, and many undifferentiated acute abdominal presentations. CTA is needed for mesenteric ischemia or major vascular concerns. Neuroimaging should be guided by time course and neurologic risk. Sudden thunderclap headache, focal neurologic deficit, trauma, anticoagulation, papilledema, immunosuppression, cancer, seizure with persistent deficit, and altered mental status with concern for structural disease warrant urgent consideration of CT or MRI.

Clinical question	Preferred imaging concept	Reason
RUQ pain after meals, fever, Murphy sign	RUQ ultrasound.	Gallstones, gallbladder wall thickening, pericholecystic fluid, bile duct dilation.
RLQ pain/appendicitis in nonpregnant adult	CT abdomen/pelvis commonly used.	High diagnostic yield and alternative diagnoses.
Pregnancy pelvic/RLQ pain	Ultrasound first; MRI if needed.	Avoid radiation when possible; evaluate ectopic/ovarian/appendix.
Pain out of proportion/AFib/lactate	CTA abdomen/pelvis.	Mesenteric ischemia is vascular diagnosis.
Thunderclap headache	Noncontrast CT early; LP/CTA depending timing.	Subarachnoid hemorrhage must be excluded.
Acute focal deficit within treatment window	Stroke protocol imaging.	Do not delay reperfusion evaluation.

6 Physical examination pearls

The physical exam is most powerful when it tests a hypothesis. A murmur is not merely “present”; its timing, location, radiation, response to maneuvers, and associated pulses/JVP change the differential. Crackles are interpreted with JVP and edema. Joint swelling is interpreted with warmth, range of motion, distribution, and systemic symptoms. Skin findings are interpreted with mucosal involvement and vital signs. Neurologic findings localize lesions before imaging. Volume status is never one sign; it is a pattern across mucous membranes, JVP, edema, orthostasis, lung findings, weight change, urine output, and ultrasound when available.

Exam domain	High-yield findings	Diagnostic connections
Volume status	JVP, edema, mucous membranes, orthostatics, crackles, skin turgor, urine output.	CHF/renal/liver overload vs dehydration/bleeding/sepsis.
Cardiac murmurs	Systolic vs diastolic, radiation, carotid pulse, maneuvers.	AS radiates to carotids; MR to axilla; HCM increases with Valsalva; diastolic murmurs are pathologic.
Lung exam	Wheeze, crackles, focal decreased sounds, egophony, dullness.	Obstruction, edema/fibrosis, effusion, consolidation, pneumothorax.
Abdomen	Peritonitis, Murphy sign, CVA tenderness, pulsatile mass, ascites.	Surgical abdomen, cholecystitis, pyelo/stone, AAA, cirrhosis/malignancy.
Neurologic localization	Cranial nerves, motor/sensory, reflexes, cerebellar, gait.	Central vs peripheral, UMN vs LMN, cortical vs brainstem vs cord.
Joint exam	Warmth, effusion, passive ROM, distribution, enthesitis, dactylitis.	Septic/crystal arthritis, RA, spondyloarthritis, OA, periarticular disease.
Skin	Primary lesion, blanching, purpura, mucosa, palms/soles, necrosis.	Infection, vasculitis, drug reaction, systemic disease.

Volume assessment is often wrong when based on a single sign. A patient with cirrhosis can have total body fluid excess and low effective arterial volume. A patient with heart failure can be intravascularly congested but hypotensive from cardiogenic shock. A patient with sepsis can have warm extremities early and cool extremities late. The most useful approach is to ask whether the problem is perfusion failure, venous congestion, interstitial edema, or fluid responsiveness. Trend weight, urine output, creatinine, sodium, JVP, edema, lung findings, and response to therapy.

7 Evidence-based medicine and statistics

Evidence-based medicine is the discipline of applying population data to an individual patient while preserving clinical judgment. A statistically significant result may be clinically trivial; a nonsignificant result may be underpowered; a relative risk reduction may sound impressive while absolute benefit is small. Screening tests require special caution because false positives, overdiagnosis, and lead-time bias can make outcomes look better without changing meaningful mortality. Treatment decisions should distinguish relative effect, absolute effect, baseline risk, harms, cost, patient values, and feasibility.

Measure	Definition	Clinical interpretation
Relative risk (RR)	Risk in exposed/treatment group divided by risk in control group.	RR less than 1 suggests reduced risk; RR greater than 1 suggests increased risk. Does not show baseline risk.
Odds ratio (OR)	Odds of event divided by odds in comparison group.	Approximates RR when events are rare; can exaggerate effect when common.
Hazard ratio (HR)	Relative event rate over time in survival analysis.	Common in time-to-event studies; assumes proportional hazards when using Cox models.
Absolute risk reduction (ARR)	Control event rate minus treatment event rate.	Directly reflects absolute benefit.
Number needed to treat (NNT)	1 / ARR.	Patients treated to prevent one event over specified time horizon.

Number needed to harm (NNH)	1 / absolute risk increase.	Patients exposed to cause one adverse event over specified time horizon.
Confidence interval	Range of values compatible with data under assumptions.	Wide CI means imprecision; crossing null often means not statistically significant.
P-value	Probability of data as or more extreme if null hypothesis true.	Does not measure effect size or probability the hypothesis is true.

Bias/confounder	Meaning	Example
Lead-time bias	Earlier diagnosis appears to prolong survival without delaying death.	Cancer screening detects disease earlier but death occurs at same time.
Length-time bias	Screening preferentially detects slower disease.	Indolent cancers overrepresented among screen-detected cases.
Overdiagnosis	Detection of disease that would never cause symptoms/death.	Low-risk tumors treated as clinically meaningful disease.
Selection bias	Groups differ systematically before exposure/outcome assessment.	Healthy volunteer effect in prevention studies.
Confounding by indication	Reason for treatment is related to outcome.	Sicker patients receive a drug and have worse outcomes independent of drug.
Immortal time bias	A period during which outcome could not occur is misclassified.	Patients must survive long enough to receive exposure.
Verification bias	Gold standard applied selectively.	Only positive screening tests get confirmatory test, inflating sensitivity/specificity.

8 Medication adverse-effect pattern recognition

Medication effects are a cross-system cause of almost every internal medicine presentation. A final review should connect medications to symptoms, lab abnormalities, and emergencies. Always ask what was started, stopped, dose-adjusted, taken over the counter, borrowed, injected, or supplemented. Older adults, renal disease, liver disease, pregnancy, polypharmacy, and drug-drug interactions increase risk. Deprescribing is a diagnostic and therapeutic intervention when the medication burden contributes to falls, delirium, hypotension, bleeding, kidney injury, or electrolyte disorders.

Medication/class	Major internal medicine patterns
ACE inhibitors/ARBs	Hyperkalemia, creatinine rise after initiation/renal artery stenosis, cough/angioedema with ACE inhibitors, fetal toxicity.
Thiazides	Hyponatremia, hypokalemia, hyperuricemia/gout, hypercalcemia, photosensitivity.
Loop diuretics	Hypokalemia, hypomagnesemia, volume depletion, ototoxicity.
NSAIDs	AKI, sodium retention/HTN/CHF worsening, GI bleed, platelet dysfunction, interstitial nephritis.
Anticoagulants/antiplatelets	GI/intracranial bleeding, anemia, peri-procedural management issues.
Steroids	Hyperglycemia, infection, osteoporosis, myopathy, mood changes, adrenal suppression, AVN.
Opioids	Respiratory depression, constipation, falls, delirium, dependence, nausea.
Benzodiazepines/Z-drugs	Falls, delirium, cognitive impairment, respiratory depression with opioids, withdrawal.
Anticholinergics	Urinary retention, constipation, dry mouth, delirium, blurred vision, tachycardia.
Statins	Myalgias, rare myopathy/rhabdomyolysis, transaminase elevations; interactions with some CYP inhibitors.
Amiodarone	Thyroid dysfunction, pulmonary toxicity, liver toxicity, corneal deposits, neuropathy, QT prolongation.
Metformin	GI effects, B12 deficiency, lactic acidosis risk in severe renal/hepatic/hypoxic states.

SGLT2 inhibitors	Genital mycotic infections, euglycemic DKA, volume depletion; sick-day holding may be needed.
Immune checkpoint inhibitors	Colitis, hepatitis, pneumonitis, endocrinopathies, myocarditis, nephritis, dermatitis.

9 Do-not-miss diagnoses and final integrated cases

A do-not-miss diagnosis is one in which delay causes death, disability, or irreversible organ injury. The diagnosis may be uncommon, but the threshold to consider it is low when the presentation has compatible red flags. Board questions often test recognition of the one feature that changes management: jaw claudication in suspected giant cell arteritis, thunderclap onset in headache, pain out of proportion in mesenteric ischemia, mucosal involvement in SJS/TEN, exertional syncope in aortic stenosis/HCM/arrhythmia, or hyperkalemic ECG changes in renal failure.

Presentation	Do-not-miss list
Chest pain	ACS, aortic dissection, PE, tension pneumothorax, tamponade, esophageal rupture.
Dyspnea	PE, pulmonary edema, pneumothorax, severe asthma/COPD, pneumonia/sepsis, anaphylaxis, ACS, neuromuscular failure.
Headache	SAH, meningitis/encephalitis, GCA, cerebral venous thrombosis, mass with increased ICP, angle-closure glaucoma, hypertensive emergency.
Back pain	Cauda equina, spinal epidural abscess, malignancy/cord compression, fracture, AAA/dissection, infection.
Abdominal pain	Mesenteric ischemia, perforation, obstruction, ectopic pregnancy, torsion, AAA, cholangitis, pancreatitis, appendicitis.
AMS	Hypoglycemia, hypoxia/hypercapnia, sepsis, stroke/ICH, seizure/nonconvulsive status, meningitis/encephalitis, intoxication/withdrawal.
Rash	SJS/TEN, meningococcemia, RMSF, toxic shock, necrotizing fasciitis, DRESS, vasculitis, anaphylaxis.
AKI	Obstruction, rapidly progressive GN, tumor lysis, rhabdomyolysis, hepatorenal syndrome, thrombotic microangiopathy.
Anemia	GI bleed, hemolysis, marrow failure/leukemia, B12 neurologic disease, pregnancy-related bleeding, malignancy.

Case	Syndrome representation	Reasoning and next steps
A 68-year-old with acute dyspnea, pleuritic chest pain, tachycardia, oxygen saturation 88 percent, and swollen left calf.	Acute hypoxemic pleuritic syndrome with VTE signs.	PE is high on differential. Assess stability. If unstable, bedside echo/urgent PE pathway. If stable, CTPA unless contraindicated; anticoagulate when probability high and bleeding risk acceptable.
A 74-year-old with confusion fluctuating over 2 days after starting diphenhydramine for sleep; afebrile, no focal deficit.	Acute fluctuating attention impairment in older adult after anticholinergic exposure.	Delirium until proven otherwise. Check glucose, oxygenation, infection/metabolic triggers, urinary retention/constipation, medication list. Stop anticholinergics.
A 52-year-old with crushing chest pressure, diaphoresis, ST elevation II/III/aVF.	Inferior STEMI.	Reperfusion pathway. Consider right-sided ECG if hypotension/clear lungs suggests RV infarct; avoid excessive nitrates if RV preload dependent.
A 35-year-old with fever, severe headache, neck stiffness, photophobia, and confusion.	Febrile meningeal encephalopathic syndrome.	Meningitis/encephalitis emergency. Blood cultures if no delay, empiric antibiotics plus or minus acyclovir; CT before LP only if safety criteria present.

A 45-year-old with fever, palpable purpura, hematuria, creatinine rise, sinus symptoms, and hemoptysis.	Pulmonary-renal vasculitic syndrome.	Pulmonary hemorrhage/RPGN emergency. Urinalysis, renal function, ANCA/anti-GBM/complements, chest imaging; urgent nephrology/rheum.
A 58-year-old with fever, RUQ pain, jaundice, hypotension, and confusion.	Severe ascending cholangitis.	Sepsis plus biliary obstruction. Fluids/vasopressors if needed, broad antibiotics, urgent ERCP/source control.
A 42-year-old with thunderclap headache during exertion, vomiting, and neck stiffness.	Sudden severe meningeal headache syndrome.	SAH until excluded. Noncontrast CT; LP/CTA depending timing and local pathway. Manage BP/pain and neurosurgical consultation if confirmed.

10 Final high-yield lab pattern table

Pattern	Think of
AST/ALT very high, often greater than 1000	Ischemic hepatitis, acetaminophen/toxin, acute viral hepatitis; check INR and mental status for acute liver failure.
Alkaline phosphatase/bilirubin predominant	Cholestasis/obstruction, infiltrative disease, PBC/PSC, drug injury; confirm hepatic source with GGT/5-nucleotidase.
Low sodium plus high urine osm plus high urine sodium	SIADH, adrenal insufficiency, hypothyroidism, renal salt wasting, diuretics; interpret with volume status.
High anion gap acidosis	Lactate, ketoacidosis, renal failure, toxic alcohols, salicylates; calculate osmolar gap if toxin concern.
Metabolic alkalosis plus hypokalemia	Vomiting/NG suction, diuretics, mineralocorticoid excess; urine chloride helps classify.
Microcytic anemia plus low ferritin	Iron deficiency until proven otherwise; search blood loss in appropriate patients.
Normocytic anemia plus high retic	Bleeding or hemolysis; check bilirubin/LDH/haptoglobin/smear.
Thrombocytopenia plus schistocytes	TMA/DIC/mechanical hemolysis; evaluate hemolysis, coagulation, renal/neuro findings.
Proteinuria plus hematuria plus RBC casts	Glomerulonephritis; nephrology evaluation.
Eosinophilia	Allergy/asthma, drug reaction, parasite, adrenal insufficiency, eosinophilic GI/lung disease, malignancy.
Low complement C3/C4	Immune-complex disease: lupus nephritis, cryoglobulinemia, postinfectious GN, endocarditis-associated GN.
Elevated ESR/CRP	Inflammation/infection/malignancy; nonspecific and must fit syndrome.

11 Capstone: how to use the whole book

The completed Medvale internal medicine book can be used in three modes. In disease mode, a learner studies an organ-system chapter and builds illness scripts. In syndrome mode, the learner starts from a chief complaint and uses this chapter to traverse possible systems. In Graph-RAG mode, each concept becomes a node: disease, symptom, lab pattern, medication, diagnostic test, complication, management step, red flag, and disposition. Edges should encode relationships such as causes, suggests, rules out, requires urgent treatment, contraindicates, mimics, complicates, treats, adverse effect of, and monitored by.

The most valuable clinical graph is not a list of facts; it is a map of decisions. For example, thunderclap headache should connect to subarachnoid hemorrhage, urgent noncontrast CT, lumbar puncture/CTA depending timing, avoid missed sentinel bleed, and neurosurgical consultation. ACE inhibitor should connect to cough, angioedema, hyperkalemia, creatinine rise, diabetic kidney protection, and contraindicated in pregnancy. Palpable purpura should connect to small-vessel vasculitis, platelet/coagulation distinction, renal urinalysis, and infection/drug/autoimmune triggers. This final chapter is designed to provide those edges.

Final review rule
When uncertain, return to first principles: stabilize the patient, localize the syndrome, compare likely and dangerous diagnoses, choose tests that change management, treat time-sensitive disease before perfect certainty, and reassess when new data contradict the initial story.

Selected source notes for this chapter

This capstone chapter is synthesized from the completed Medvale organ-system sequence and from the uploaded NBME internal medicine content outline and Step-Up table of contents. The NBME outline emphasizes diagnosis and management tasks and includes major internal medicine systems, emergency/inpatient care, and older adult care; the Step-Up outline includes ambulatory medicine and appendix topics such as radiographic interpretation, ECG interpretation, physical examination pearls, common problem workups, statistics/evidence-based medicine, and end-of-life issues. These source structures guided the chapter organization, but the prose and algorithms are original synthesis.